Solution:

Let MD denote the event that the monthly Mass Drill programme is implemented in the school. Also, let

A: Mr.Arivazhagan is appointed

B: Mr.Ilavarasan is appointed

C: Mr.Anbarasan is appointed.S

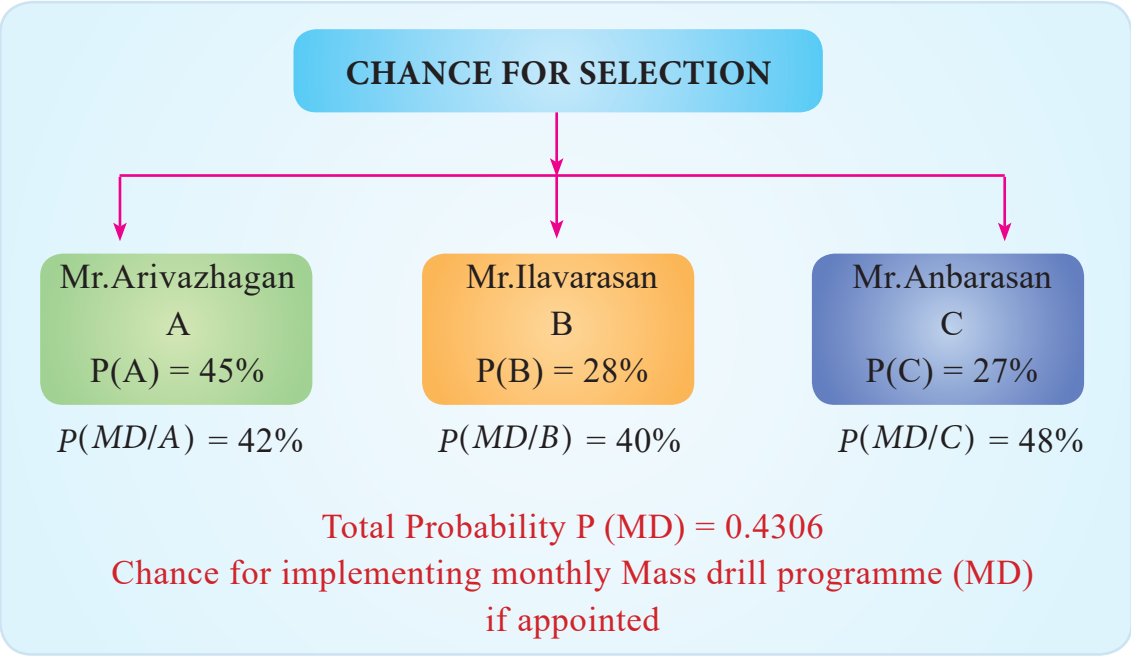


Fig. 8.10 Appointment of Physical Teacher

From fig. 8.10 the information given about these events are

$P(A) = 0.45$ $P(B) = 0.28$ $P(C) = 0.27$

$P(MD/A) = 0.42$ $P(MD/B) = 0.40$ $P(MD/C) = 0.48$

With these, the probability for implementing monthly Mass Drill programme in the school can be computed using total probability as

$$P(MD) = P(MD/A)P(A) + P(MD/B)P(B) + P(MD/C)P(C)$$

$$= (0.42 \times 0.45) + (0.40 \times 0.28) + (0.48 \times 0.27)$$

$$= 0.189 + 0.112 + 0.1296$$

$$\therefore \quad P(MD) = 0.4306.$$

If it is known that the monthly Mass Drill programme is implemented in the school,